PLANT DOWNTIME & OPERATIONAL EFFICIENCY ANALYSIS

Meridian Steelworks | Tableau Dashboard Project

1. Project Overview

This project analyses the production performance and operational efficiency of Meridian Steelworks. The objective was to evaluate machine performance, identify production losses, examine downtime, assess product quality, and compare actual production with expected production. Excel was used for initial data inspection, while Tableau was used for KPI development, analysis, and interactive visualisation.

2. Data Inspection and Preparation

The raw dataset was first inspected in Excel to understand its structure and identify missing values. During inspection, 12 records were found to contain unknown values across all columns. These records had no meaningful information and did not contribute to the production analysis.

The production trend also contained a null date category associated with 29,101 units. Since the dates for these records were not recorded, the units could not be assigned to specific months.

3. Key Performance Indicators (KPIs)

The following KPIs were developed to measure production efficiency, machine availability, performance, product quality, downtime, and energy consumption.

KPI	Formula / calculation
Overall Equipment Effectiveness (OEE)	Availability × Performance × Quality
Availability %	Operating Minutes ÷ Planned Minutes × 100
Performance %	Actual Units Produced ÷ Expected Units Production × 100
Quality %	Good Units Produced ÷ Actual Units Produced × 100
Total Downtime Minutes	Sum of Downtime Minutes
Planned Minutes	Sum of Planned Minutes
Operating Minutes	Planned Minutes – Downtime Minutes
Total Machines	Count of distinct machines
Actual Units Produced	Sum of Units Produced
Expected Units Production	Ideal Rate per Minute × Operating Minutes
Total Units Rejected	Sum of Units Rejected
Reject Rate %	Total Units Rejected ÷ Actual Units Produced × 100
Units Rejected per Run	Total Units Rejected ÷ Number of Production Runs
Production Shortfall %	(Expected Production – Actual Production) ÷ Expected Production × 100
Total Energy Consumption	Sum of Energy Consumption (kWh)
Energy per Unit	Total Energy Consumption ÷ Actual Units Produced

4. Key Findings

Dashboard 1: Plant Production Performance Overview

OEE by production line. Line B recorded the highest OEE at 69.33%, followed by Line A at 68.33%. Line C recorded the lowest at 67.24%. The difference between the lines was relatively small.

Production trend by date. The null date category accounted for 29,101 units, which could not be assigned to a specific month. April recorded the highest monthly production at approximately 130,733 units, followed by June at approximately 129,640 units. Production dropped to 61,002 units in July and remained relatively low from July to December, ranging between approximately 61,002 and 78,082 units.

Actual vs. expected production by shift. Morning produced 398,193 units against 476,240 expected; Night produced 385,480 against 465,345; and Afternoon produced 370,902 against 441,930. Afternoon had the lowest actual production, followed by Night, while Morning had the highest. All three shifts fell below expected production.

OEE by machine. M01 recorded the highest OEE at 71.51%, while M06 recorded the lowest at 64.85%.

Dashboard 2: Machine Efficiency Overview

Actual vs. expected production by machine. Every machine produced below its expected output. M02 recorded the highest actual production at 109,699 units, while M08 recorded the lowest at 82,075 units.

Downtime by machine. M02 recorded the highest downtime at 8,378 minutes, while M08 recorded the lowest at 5,900 minutes.

Production shortfall by machine. M08 had the highest shortfall at 20.92%. M01 had the lowest at 12.89%.

Dashboard 3: Production Quality & Performance Analysis

Machine–product rejection rates. M09 producing Sheet 4mm recorded the highest rejection rate at 19.98%. M02 producing Rebar 16mm recorded the lowest rejection rate among the combinations examined, at 3.14%. Rejection rates therefore varied across machine–product combinations.

Energy consumption by machine. M01 recorded the highest energy consumption at approximately 140,221 kWh, while M09 recorded the lowest at approximately 107,328 kWh. Energy usage should be considered alongside production output when assessing machine efficiency.

Machine performance summary. M01 on Line A recorded the highest OEE at 71.51%, while M06 on Line A recorded the lowest at 64.85%.

5. Overall Analysis

The plant recorded an overall OEE of **68.56%**, with Availability at **87.4%**, Performance at **83.5%**, and Quality at **94.05%**. Quality was the strongest OEE component, while Performance was the weakest.

Actual production was consistently lower than expected across all machines and shifts. The overall production shortfall was approximately **16.55%**. The plant recorded over **84,000 minutes of downtime**. Operating time was **580,539 minutes**, compared with **664,560 planned minutes**.

The plant also recorded **68,721 rejected units**, an overall rejection rate of **5.95%**, and an average of approximately **49 rejected units per production run**. These findings show that production losses, machine downtime, and product rejection are important factors affecting operational efficiency.

6. Recommendations

1. Reduce machine downtime. Investigate the causes of downtime, particularly on M02, which recorded the highest downtime. Introduce preventive maintenance schedules and ensure machines are regularly inspected and maintained.

2. Improve production performance. Since all machines are producing below expected output, investigate the production gap. Areas to examine include machine operating conditions, production processes, operator efficiency, and interruptions such as power supply issues.

3. Prioritise underperforming machines. M06 recorded the lowest OEE, while M08 had the highest production shortfall. Prioritise these machines to identify and address the factors affecting their performance.

4. Improve shift productivity. The Afternoon shift recorded the lowest production. Review its operating conditions, staffing, machine availability, and production processes.

5. Address product rejection. M09 producing Sheet 4mm had the highest rejection rate. Review this machine–product combination to identify causes of defects and reduce material waste.

6. Monitor energy efficiency. M01 recorded the highest energy consumption. Evaluate energy usage alongside production output to understand whether machines are consuming energy efficiently.

7. Improve production record-keeping. Ensure production dates and other operational information are properly recorded. The 12 completely unknown records and records with missing production dates highlight the importance of accurate data collection.

7. Conclusion

The analysis shows that Meridian Steelworks is operating below its expected production capacity. Although the plant recorded an overall OEE of 68.56% and a quality score of 94.05%, its performance is affected by production shortfalls, substantial downtime, and product rejection. The most important priorities are improving production performance and reducing downtime across machines and shifts. Addressing these issues can help improve output, reduce operational losses, minimise rejected products, and make better use of production resources.